BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Testing: Normal, Boundary and Invalid Data

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why use meaningful variable names?

2. What is FTP used for?

A) Transferring files
B) Sending email
C) Looking up domain names
D) Assigning IP addresses

3. "Malware means any program that uses a large amount of RAM." Correct this. Explain why.

4. A school website is flooded with requests. "A denial-of-service attack makes access better for real users." Correct this. What happens to them?

Today's objectives

Choose suitable data and explain iterative and final testing. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Normal test data

Representative valid inputs normally expected during use.

Boundary test data

Values at, immediately below and immediately above an allowed limit.

Invalid or erroneous

Inputs rejected by the required rules, including out-of-range or wrong-format data.

Iterative and final testing

Iterative testing runs during development; terminal testing checks the finished solution.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Normal test data	Representative valid inputs normally expected during use. Consider this in a test a marks form.
Boundary test data	Values at, immediately below and immediately above an allowed limit. Consider this in a ticket quantity checker.
Invalid or erroneous	Inputs rejected by the required rules, including out-of-range or wrong-format data. Consider this in a age-restricted service.
Iterative and final testing	Iterative testing runs during development; terminal testing checks the finished solution. Consider this in a sports score input.

I Do - One worked example

A test a marks form needs normal test data and boundary test data. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Normal test data [1]; Representative valid inputs normally expected during use. [1].

3. Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1].

4. Give a specific point and a linked reason for each feature.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a ticket quantity checker, explain the purpose of normal test data. [2 marks]

2. A age-restricted service needs boundary test data. Explain its role. [2 marks]

3. A sports score input uses invalid or erroneous and iterative and final testing. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the test value to the highest valid unsigned 8-bit integer.

```
value = 256
if value >= 0 AND value <= 255 then
print("Accepted")
else
print("Rejected")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how boundary test data and invalid or erroneous help in a club attendance tool. [4 marks]

Q4. Explain the roles of normal test data, invalid or erroneous and iterative and final testing in a online purchase form. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Normal test data contains valid inputs that are typically expected during use. ☐ **T** ☐ **F**
2. Representative valid inputs normally expected during use ☐ **T** ☐ **F**
3. Values at, immediately below and immediately above an allowed limit ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Clear names show what values mean and help people read and change code. ☐

Q2 A - Transferring files ☐

Q3 Malware is made to cause harm, disruption or access without permission. Different types work in different ways. ☐

Q4 A denial-of-service attack floods a service with traffic or requests. Genuine users then cannot use it. ☐

Code First - Model

```
value = 255
if value >= 0 AND value <= 255 then
print("Accepted")
else
print("Rejected")
endif
```

Output: Accepted

We Do - Check the model

1. Normal test data [1]; Representative valid inputs normally expected during use. [1]. ☐

2. Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1]. ☐

3. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. Iterative and final testing [1]; Iterative testing runs during development; terminal testing checks the finished solution. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Normal test data [1]; Invalid or erroneous [1]. ☐

Q3 Boundary test data [1]; Values at, immediately below and immediately above an allowed limit. [1]. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. ☐

Q4 Normal test data [1]; Representative valid inputs normally expected during use. [1]. Invalid or erroneous [1]; Inputs rejected by the required rules, including out-of-range or wrong-format data. [1]. Iterative and final testing [1]; Iterative testing runs during development; terminal testing checks the finished solution. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: